

Phoebe Sandhaus

Astrophysics Researcher and educator

CONTACT

phoebe.sandhaus@gmail.com

(410) 370-5540

https://phoebesandhaus.github.io

github.com/PhoebeSandhaus

orcid.org/0000-0002-6019-4818

Troy, NY

EDUCATION

ASTRONOMY AND ASTROPHYSICS, PH.D.

Pennsylvania State University

2027

ASTRONOMY AND ASTROPHYSICS, M.S.

Pennsylvania State University

2022

PHYSICS, B.S.

University of Maryland, Baltimore County

2019

PROFESSIONAL SUMMARY

Currently, a PhD Candidate in Astronomy and Astrophysics at Penn State, focusing on exoplanet modeling and planet formation simulations. Passionate advocate for increasing accessibility of science. Looking to join an environment that values students, teaching, and service, just as much as novel research.

TEACHING EXPERIENCE

INSTRUCTOR, ASTRO 297: ASTRONOMY COMMUNICATION

Pennsylvania State University, Department of Astronomy and Astrophysics

Fall 2022

- Course covered verbal and written communication skills, how to apply these skills to a technical topic within astronomy
- Updated syllabus to include new aspect of final grade called “professionalism”
- Created associated detailed rubrics for each assignment to set clear expectations for students

GRADUATE TEACHING ASSISTANT

Pennsylvania State University, Department of Astronomy and Astrophysics

2020 - Current

- Assisted in grading and teaching an incredibly wide range of courses over 13+ semesters: large (>300 students) introductory level courses to small (<15 students) upper-level (including a graduate course) courses
- Ran weekly online planetarium shows using the program Stellarium between Fall 2020-Spring 2021

GUEST LECTURER, PHYS 1961: PROFESSIONAL DEVELOPMENT FOR PHYSICISTS

Rensselaer Polytechnic Institute, Department of Physics, Applied Physics, and Astronomy

Fall 2023

- “Science, Objectivity, and Diversity”: discussed how scientific research can be influenced by beliefs and biases of scientists
- Divided students into groups to discuss open-ended questions about the topic

GUEST LECTURER, ASTRO 5: THE SKY AND PLANETS

Pennsylvania State University, Department of Astronomy and Astrophysics

Fall 2020

- “SETI: The Search for Extraterrestrial Intelligence”: lectured on the history and current state of the field of SETI

RESEARCH PROJECTS

EXOZIPPY: RECREATING POPULAR EXOPLANET MODELING SOFTWARE SUITE TO INCREASE ACCESSIBILITY

Primary Advisor: Jason Eastman, Harvard-Smithsonian Center for Astrophysics

2024 - Current

- Prominent exoplanet modeling code EXOFASTv2 currently is written in the proprietary coding language IDL which presents significant barrier to the community
- Rebuilding code from the ground up in Python to increase accessibility and community involvement
- Preliminary benchmarking shows consistent results, with *at least* an order of magnitude speed increase
- Code publicly available on GitHub: https://github.com/jdeast/EXOZIPPy

SUPER-EARTH FORMATION: N-BODY SIMULATIONS AND ORBITAL DYNAMICS

Primary Advisor: Mariah MacDonald, The College of New Jersey

2020 - Current

- Investigated how different initial conditions in planet formation simulations affect the properties of formed inner planets
- Gained proficiency in Unix shell scripting and requesting computing resources through Slurm job scripts
- Wrote program for the Python package Reboundx that calculates gas damping forces on bodies during *N*-body simulations which is publicly available on GitHub

SEARCHING FOR PLANETS AROUND WHITE DWARFS

Primary Advisor: John H. Debes, Space Telescope Science Institute

2013-2016

- Analyzed archival white dwarf data from the Cosmic Origins Spectrograph (COS) on the Hubble Space Telescope
- Converted spectrographic data into time-series photometry to search for short-term time variability that could be evidence of an object transiting (e.g., a planet)
- Performed injection recovery tests to determine that we would have been able to detect transiting sub-Earth-sized objects in orbit around our sample of white dwarfs

SERVICE & OUTREACH

MEMBER OF CLIMATE & DIVERSITY COMMITTEE

Department of Astronomy and Astrophysics, Pennsylvania State University

2020 - Current

- Created an initiative called the "Astro+ Research Showcase" in which we would put on a presentation to introduce and connect undergraduates to research opportunities at Penn State; I have organized a total of 10 showcases, and designed organizing materials such that the events can continue in perpetuity without me
- Organized and ran three undergraduate town halls; wrote a report summarizing the comments and turned the comments into actionable initiatives (e.g., Astro+ Research Showcase)
- Organized an event for people to write biographies of scientists with underrepresented minoritized identities that were made into posters to display in the department

MEMBER OF UNDERGRADUATE CURRICULUM COMMITTEE

Department of Astronomy and Astrophysics, Pennsylvania State University

2021 - 2025

- Contributed significantly to the curricular development of a new major "Astronomy" that is going through the final stages of approval
- Helped to develop the STAR (Students Together for Astronomy Research) Program (https://sites.psu.edu/astronomyresearch/), in which undergraduates could take part in a semester-long research project for course credit; Developed a detailed grading rubric/ program outline to be used by the mentors and the directors of STAR to clearly outline expectations to students
- Wrote a detailed document of recommendations for changes to the undergraduate curriculum based on DFW rates, undergraduate town hall reports, and recommendations from the Astronomy Decadal Survey

MEMBER OF GRADUATE PROGRAM COMMITTEE

Department of Astronomy and Astrophysics, Pennsylvania State University

2020 - 2021

- ·Helped to organize survey to poll graduate students' opinions and suggestions surrounding the existing qualifying exam
- Wrote detailed document of recommendations for changes to the qualifying exam based on survey results and presented findings to the larger committee

OUTREACH

Pen Pal, Letters to a Pre-Scientist	2025-2026, 2024-2025
Invited Speaker, Central Pennsylvania Observers, Cherry Springs, PA	Sep 2024
Invited Speaker, STEM Research & Innovation Symposium, Baltimore, MD	May 2024
Invited Speaker, STEM Student Leadership Conference, Baltimore, MD	May 2024
Speaker, Astronomy on Tap, State College, PA	Mar 2023
Invited Speaker, Central Pennsylvania Observers, Cherry Springs, PA	Sep 2022
Invited Speaker, Central Pennsylvania Observers, State College, PA	Apr 2022
Speaker, Astronomy on Tap, State College, PA	Aug 2021

HONORS & AWARDS

Dean's Climate and Diversity Group Award – Department of Astronomy and Astrophysics Climate and Diversity Committee, Eberly College of Science, Pennsylvania State University	2025
Honorable Mention, Chambliss Astronomy Achievement Student Award, 245th Meeting of the American Astronomical Society	2025
Zaccheus Daniel Award, Pennsylvania State University	2023
Climate Award, Department of Astronomy and Astrophysics, Pennsylvania State University	2022
Homer F. Braddock Scholarship in Biology, Chemistry, and Physics, Eberly College of Science, Pennsylvania State University	2020
University Graduate Fellowship, Pennsylvania State University	2019
Cecilia Payne-Gaposchkin Science Achievement Graduate Scholarship in Astronomy, Eberly College of Science, Pennsylvania State University	2019
College of Natural and Mathematical Sciences Scholars Program, University of Maryland, Baltimore County	2018
Outstanding Undergraduate Learning Assistant in Physics, Department of Physics, University of Maryland, Baltimore County	2018
Outstanding Undergraduate Learning Assistant in Physics, Department of Physics, University of Maryland, Baltimore County	2017

SCIENTIFIC PRESENTATIONS

CONTRIBUTED CONFERENCE TALKS

Simulating the Effects of Outer Giant Planets on Inner Small Planets with In Situ Formation Models, 9th Emerging Researchers in Exoplanet Science Symposium, Ithaca, NY	Jul 2024
Simulating the Effects of Outer Giant Planets on Inner Super-Earths with In Situ Formation Models, 55th Annual Meeting of the Division on Dynamical Astronomy, Toronto, Canada	May 2024
Violent Outbursts of Young Stars: UV Analysis of FUor HBC 722, Space Astronomy Summer Program Student Presentations, Space Telescope Science Institute, Baltimore, MD	Aug 2017
A Search for Short-period Rocky Planets around WDs with COS, National Capital Area Disks Meeting 6, Washington, D.C.	Jul 2016

CONTRIBUTED CONFERENCE POSTERS

EXOZIPPy: A Python Adaptation of the Exoplanet Modeling Code EXOFASTv2, Astrobiology Graduate Conference, Boulder, CO	Jun 2025
EXOZIPPy: A Python translation of EXOFASTv2 to simultaneously model stars and planets, 245th Meeting of the American Astronomical Society, National Harbor, MD	Jan 2025
EXOZIPPy: A Python translation of EXOFASTv2 to simultaneously model stars and planets, 55th Annual Meeting of the Division on Dynamical Astronomy, Toronto, Canada	May 2024
Simulating the Effects of Outer Giant Planets on Inner Super-Earths with In Situ Formation Models, 54th Annual Meeting of the Division on Dynamical Astronomy, Lansing, MI	May 2023
Simulating the Effects of Outer Giant Planets on Inner Small Planets with In Situ Formation Models, 6th Emerging Researchers in Exoplanet Science Symposium, Virtual	May 2021
Simulating the Effects of Outer Giant Planets on Inner Super-Earths with In Situ Formation Models, 52nd Annual Meeting of the Division on Dynamical Astronomy, Virtual	May 2021
A Search for Rocky Planets in Close-in Orbits Around White Dwarfs, 227th Meeting of the American Astronomical Society, Kissimmee, FL	Jan 2016

INVITED COLLOQUIA, SEMINARS, & LUNCH TALKS

Simulating the Effects of Outer Giant Planets on Inner Super-Earths with In Situ Formation Models, Exoplanets, Star and Planet Formation Seminar Series, Space Telescope Science Institute, Virtual	Nov 2021
Simulating the Effects of Outer Giant Planets on Inner Super-Earths with In Situ Formation Models, Department of Astronomy and Astrophysics Lunch Talk Series, Pennsylvania State University, Virtual	Apr 2021
Forming Diverse Super-Earth Systems In Situ (MacDonald et al. 2020), Department of Astronomy and Astrophysics Lunch Talk Series, Pennsylvania State University, Virtual	Oct 2020
Uncertainty: Where does it come from and how do we represent it in our data?, Measurement Bootcamp, University of Maryland, Baltimore County Joint Center for Earth System Technology	Oct 2018

PUBLICATIONS

First Author

Sandhaus, P., Eastman, J. et al. "EXOZIPPy: A Python translation of EXOFASTv2 to simultaneously model stars and planets," in prep.

Sandhaus, P., Dawson, R. I., MacDonald, M. G., Shakespeare, C. J., and Morrison, S. "Effects of Outer Giant Planets on In Situ Formation of Inner Super-Earths," The Astrophysical Journal, Volume 990, Issue 1, (2025).

Sandhaus, P. H., Debes, J. H., Ely, J., Hines, D. C., and Bourque, M., "A Search for Short-Period Rocky Planets Around WDs with the Cosmic Origins Spectrograph (COS)," The Astrophysical Journal, Volume 823, Issue 1, (2016).

Contributing Author

Rodriguez, J., **Sandhaus, P.**, et al. in prep.

MacDonald, M. G., **Sandhaus, P.**, and Brefka, L. "Small Planets Forming in the Inner Disk Do Not Need Migration," Research Notes of the AAS, in review.

MacDonald, M. G., Brefka, L., **Sandhaus, P.**, Pojar, K., and Zeringue, M. "Resonance is Not More Prevalent Among Young, Close-in Planets," The Astrophysical Journal, in review.

Brefka, L., Dawson, R. I., MacDonald, M. G., **Sandhaus, P.**, and Ford, E. B. "Effects of pebble accretion isolation mass on observable exoplanet properties," Monthly Notices of the Royal Astronomical Society, Volume 552, Issue 1, (2026).

Tusay, N. and Huston, M. J., et al. (including **Sandhaus, P.**). "A Search for Radio Technosignatures at the Solar Gravitational Lens Targeting Alpha Centauri," The Astronomical Journal, Volume 164, Issue 3, (2022).

REFERENCES

JASON EASTMAN

Harvard College Observatory Research Associate,
Harvard-Smithsonian Center for Astrophysics

Phone: (617) 495-3752

Email: jason.eastman@cfa.harvard.edu

MARIAH G. MACDONALD

Clare Boothe Luce Assistant Professor of Physics,
The College of New Jersey

Phone: (509) 481-0459

Email: mmacdonam@tcnj.edu

JAMES H. TUTT

Research Professor of Astronomy and Astrophysics,
Pennsylvania State University

Phone: (814) 865-6338

Email: jht12@psu.edu

JASON T. WRIGHT

Professor of Astronomy and Astrophysics,
Pennsylvania State University

Email: astrowright@gmail.com